

RAUC dm-verity PWU-AB-2 — Install attempt + RED_MODE=1 baseline

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1. rauc CLI installation on macOS host

`brew install rauc` — **FAILED**. No Homebrew formula exists for rauc (suggestion was runc, which is unrelated).

`rauc build` from source (<https://github.com/rauc/rauc>) — **FAILED on macOS**: - `meson setup build` failed with: Dependency "libnl-genl-3.0" not found - libnl is a Linux-specific netlink library; the Homebrew libnl formula exists but does NOT install a `libnl-genl-3.0.pc` pkg-config file on macOS (macOS has no netlink). - **Root cause**: rauc depends on Linux kernel interfaces (netlink/genl, dbus, fdisk) and cannot build natively on macOS.

Conclusion: rauc host binary is not buildable on macOS. The project’s CI/build infrastructure (inside the Podman aarch64 Linux container where Buildroot builds) or a dedicated Linux host is needed for bundle building.

`u-boot-tools` (`fw_setenv/fw_printenv`):

Package	Status
u-boot-tools	Installed via <code>brew install u-boot-tools</code> (v2026.04)

What works on macOS host

All the tools needed to run the test driver itself are available: - `qemu-system-aarch64` — installed (QEMU 11.0.1) - `expect` — available at `/usr/bin/expect` - `openssl` — available (key generation)

2. Dev keys generation

PASS. `bash tests/emulator/ab_virt/rauc/gen_dev_keys.sh` succeeded:

Artifact	Path
Private key	<code>tests/emulator/ab_virt/out/rauc-keys/dev.key.pem</code>

Artifact	Path
Public cert	tests/emulator/ab_virt/out/rauc-keys/dev.cert.pem
Fingerprint	CB:2D:F8:80:75:13:B9:46:A2:AC:5A:3D:89:02:E1:D0:78:61:20:01:2A:27:7E:3

3. Build artifacts

Preconditions met — all artifacts present and intact:

Artifact	Size	Status
out/.ok	42 B	Present
out/.disk_ok	51 B	Present
out/images/u-boot.bin	1.0 MB	U-Boot 2024.01 (Jun 11 2026)
out/images/Image	11 MB	Linux 6.1.44
out/images/rootfs.ext2	60 MB	Buildroot ext4
out/images/ab_disk.img	1.0 GB	GPT with vda1 (fat boot), vda2 (slot A root), vda3 (slot B root)
out/images/boot.scr	5.8 KB	U-Boot boot script

4. ab_rauc_verity.sh RED_MODE=1 run

QEMU acceleration workaround

QEMU 11.0.1 on Apple Silicon has a serial-output issue with `-accel hvf` — the UART produces no output and expect times out waiting for the autoboot prompt. Prior runs (PWU-AB-1/AB-3) which also used HVF had the same issue but were run with an older QEMU or different configuration.

Workaround applied: The script was extended with `QEMU_ACCEL` environment variable support (default `hvf`). The run used `QEMU_ACCEL=tcg` which works correctly (serial output intact, though ~30x slower).

Run details

Field	Value
Run ID	20260618T210924Z-ab-rauc-verity
RED_MODE	1 (defect-present baseline)
Acceleration	tcg

Field	Value
CPU	max
Guest root PW	helixota
Bundle path (guest)	/root/update.raucb (does NOT exist — expected)

Console evidence

540 lines captured at /Volumes/T7/Projects/helix_ota/docs/qa/20260618T210924Z-ab-rauc-verity/console.log

Verdict: PASS (RED baseline)

Exit code: 0 expect rc: 0

All 6 assertions correct:

#	Assertion	Expected (RED_MODE=1)	Actual	Result
1	First boot slot A	PASS (baseline must hold)	HELIX_PRE SLOT=A	PASS
2	rauc install driven	PASS (binary invoked)	Failed to initialize context	PASS
3	Post-reboot shell live	PASS (post-login sentinel)	HELIX_DONE_RAUC_MARK	PASS
4	rauc install rc=0	Must FAIL (no bundle)	HELIX_RAUC_INSTALL_RC=1	PASS (expected fail)
5	dm-verity active	Must FAIL (not wired)	HELIX_DMVERITY=0	PASS (expected fail)
6	Slot switched to B	Must FAIL (apply unproven)	HELIX_POST SLOT=A	PASS (expected fail)

5. Analysis of RED baseline findings

The test correctly proves the defect-present state. Here is what the RED baseline reveals about what is missing for a GREEN RED_MODE=0 run:

Blocker 1: /etc/rauc/system.conf not installed in rootfs (HIGHEST PRIORITY)

rauc status and rauc install both fail with:

Failed to initialize context: Failed to load system config (/etc/rauc/system.conf):
Failed to open file ?/etc/rauc/system.conf?: No such file or directory

The system.conf at tests/emulator/ab_virt/rauc/system.conf exists in the repo but is NOT wired into the Buildroot rootfs overlay. **Fix:** Add a Buildroot rootfs overlay (board/helix/ab_virt/rootfs_overlay/etc/rauc/system.conf) in build_image.sh that copies the config artifact and dev cert into both slot root filesystems.

Blocker 2: No dm-verity userspace tools in rootfs

dmsetup: not found — the device-mapper userspace tools are missing from the Buildroot config. While the kernel has dm-verity support compiled in, the userspace dmsetup binary is needed to probe verity status. **Fix:** Add BR2_PACKAGE_LVM2=y (provides dmsetup) to the Buildroot config in build_image.sh.

Blocker 3: No fw_env.config — fw_setenv fails

HELIX_FWSET_RC=1 — fw_setenv cannot operate because /etc/fw_env.config does not exist in the guest. The fw_env.config at tests/emulator/ab_virt/rauc/fw_env.config exists in the repo but is commented out and not installed. **Fix:** (a) Determine the actual U-Boot env storage config for the qemu_arm64 defconfig, (b) uncomment/correct the offset/size values in fw_env.config, (c) install it via rootfs overlay.

Blocker 4: No RAUC bundle (update.raucb) exists

rauc install cannot operate because no bundle was built. **Fix:** Requires rauc CLI on a Linux host (Podman container or CI) to run rauc bundle with the dev key+cert and a fresh rootfs image.

6. Next steps to flip GREEN (RED_MODE=0)

Ordered by dependency:

1. **Build rauc inside the Podman aarch64 container** (where Buildroot builds) and use it to call rauc bundle with the dev keys from out/rauc-keys/ and the rendered manifest.raucm.in.
2. **Add BR2_PACKAGE_LVM2=y** to the Buildroot config to get dmsetup in the rootfs.
3. **Add rootfs overlay** wiring system.conf + fw_env.config + dev.cert.pem into /etc/rauc/ and /etc/ for BOTH slots.
4. **Round-trip-verify fw_setenv/fw_printenv** against U-Boot's env storage.
5. **Rebuild** guest image with the overlay, **re-run** ab_rauc_verity.sh with RED_MODE=0 to prove slot switch + dm-verity active.

Canonical authority for the bundle-build procedure: tests/emulator/ab_virt/rauc/README.md + docs/design/rk3588_ab_virt/PWU_AB_2_RAUC_VERITY.md.